MILITARY SPECIFICATION SHEET

PRINTED CIRCUIT CARD, M1 AND M1A1 TANK DRIVER'S INSTRUMENT PANEL, DIP-A3

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the Driver's Instrument Panel, DIP-A3, described herein shall consist of this document and the latest issue in effect of specification MIL-P-62341.

3. REQUIREMENTS

3.4 Design and construction. The printed circuit card (PCC) DIP-A3 maintains sufficient current flow through malfunction sensors to prevent oxidation of sensor contacts. DIP-A3 contains pullup necessary for the logic circuitry on the DIP-A2 circuit assembly. In addition, DIP-A3 incorporates a 5 second time delay and latching circuit to eliminate nuisance indications caused by differential pressure sensors.

3.4.1 Configuration.

3.4.1.1 Class 1. Printed circuit card DIP-A3 shall be fabricated and assembled in accordance with Drawing 12282631.

3.4.1.2 Class 2. Printed circuit card DIP-A3 shall be fabricated and assembled in accordance with Drawing 12316386.

3.4.2 Weighs. The weight of the PCC shall not exceed 1.5 pounds (lb) (see 4.8.3).

3.4.4 Nuclear hardening. - Not applicable.

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3.5.1 Circuit continuity. This characteristic is not specified.

3.5.2 Circuit isolation. This characteristic is not specified.

3.5.3 Diode characteristics.

3.5.3.1 Class 1.

3.5.3.1.1 Diode circuits. Circuits which include diodes shall perform so that when 0.5 ± 0.05 amperes (A) is applied to each pin (positive) listed in table I, the voltage from each pin (positive) to each pin (ground) shall be 0.7 to 1.3 volts (V) direct current (dc) (see 4.8.4.4.1).

TABLE I. Diode circuits.

Positive	Ground
P1-1	P1-17
P1-2	P1-17
P1-4	P1-17
P1-6	P1-17
P1-8	P1-17
P1-11	P1-17
P1-12	P1-17
P1-16	P1-17
P1-20	P1-17
P1-23	P1-17
P1-24	P1-17
P1-25	P1-17
P1-28	P1-17
P1-29	P1-17
P1-31	P1-17
P1-33	P1-17

3.5.3.1.2 Reverse leakage current. The leakage current in the reverse direction shall not exceed 25 microampere at 50.0 ± 0.5 V dc (see 4.8.4.4.2).

3.5.3.2 Class 2.

3.5.3.2.1 Diode circuit characteristics. circuits which include diodes shall perform so that when 10.0 ± 0.05 milliamperes (mA) is applied to each pin (positive) listed in table II, the voltage from each pin (positive) to each pin (ground) shall be 1.4 to 0.10 V dc (see 4.8.4.4.1).

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TABLE II. Diode circuits.

Positive	Ground
P1-1	P1-17
P1-2	P1-5
P1-4	P1-7
P1-6	P1-17
P1-8	P1-17
P1-11	P1-17
P1-16	P1-17
P1-20	P1-17
P1-23	P1-17
P1-24	P1-17
P1-25	P1-17
P1-28	P1-17
P1-29	P1-17
P1-31	P1-17
P1-33	P1-17

3.5.4 Blocking diodes.

3.5.4.1 Class 1. With a 24 ± 6 V dc power source applied through an MS25237-387 lamp to P1-2 and P1-5 (positive), only the lamp at P1-2 shall illuminate when the power source return connection is made to P1-17 (ground) (see 4.8.4.5).

3.5.4.2 Class 2. With a 24 ± 6 V dc power source applied through an MS25237-387 test lamp to P1-5, a second MS25237-387 test lamp applied through the same source to P1-7, and the power source return connected to P1-17, the two test lamps shall not illuminate (see 4.8.4.5).

3.5.5 Resistors. With 24 ± 1.0 V dc applied to P1-19 (positive), the current through each pin (ground) specified in table III and IV shall be 37.0 ± 7.0 mA (see 4.8.4.6).

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3.5.5.1 Class 1

TABLE III. Resistor circuites.

Positive	Ground
P1-19	P1-3
P1-19	P1-5
P1-19	P1-7
P1-19	P1-9
P1-19	P1-10
P1-19	P1-13
P1-19	P1-14
P1-19	P1-18
P1-19	P1-21
P1-19	P1-22
P1-19	P1-26
P1-19	P1-27
P1-19	P1-30
P1-19	P1-32
P1-19	P1-34
P1-19	P1-35

3.5.5.2 Class 2.

TABLE IV. Resistor circuits.

Positive	Ground
P1-19	P1-3
P1-19	P1-5
P1-19	P1-7
P1-19	P1-9
P1-19	P1-10
P1-19	P1-13
P1-19	P1-18
P1-19	P1-21
P1-19	P1-22
P1-19	P1-26
P1-19	P1-27
P1-19	P1-30
P1-19	P1-32
P1-19	P1-34
P1-19	P1-35

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3.5.6 Relay operation. Not applicable.

3.5.7 Transient voltage characteristics. With 1.0 ± 0.5 mA applied thorugh a 2.4 K ± 10% ohm resistor to P1-17 (positive) and P1-15 (ground), the voltage from pin P1-17 (positive) to P1-15 (ground) shall be 39 ± 3 V dc (see 4.8.4.8).

3.5.8 Time delay and latching circuit.

3.5.8.1 Class 2. With a 24 ± 6 V dc power source applied through an MS25237-387 test lamp to P1-28, and through a second MS25237-387 test lamp to P1-12, and with the 24 ± 6 V dc power source connected to P1-19 and the power source return connected to P1-15 (for a 10 second period), the two test lamPs shall not Illuminate. With the power source connected to P1-27 (for a 5 ± 1 second period) and the power source return connected to P1-15, both test lamps shall illuminate and remain illuminated. Momentary interruption of the 24 ± 6 V dc power source to P1-19 shall cause both test lamps to extinguish (see 4.8.4.10).

3.5.9 Fire warning circuit. Not applicable.

4.8 Methods of inspection.

4.8.3 Weight. Conformance to 3.4.2 shall be determined by weighing the printed circuit card on a scale.

4.8.4.4 Diode circuit characteristics.

4.8.4.4.1 Diode circuits. To determine conformance to 3.5.3.1.1 and 3.5.3.2.1, utilizing a power supply of 0.5 ± 0.05 A individually to each of the positive pins of tables I or II. With the multlmeter verify that the voltage measured between each positive pin and P1-17 (ground) is 0.7 to 1.3 v dc.

4.8.4.4.2 Reverse leakage current (Class 1). To determine conformance to 3.5.3.1.2, a multimeter shall be used to measure leakage current in the reverse direction.

4.8.4.5 Blocking diodes. To determine conformance to 3.5.4.1 and 3.5.4.2, utilizing a power supply apply 24 + 6 V dc through an MS25237-387 lamp to (each) positive pins P1-2 and P1-5. Connect the power source return to P1-17 (ground) and verify that only the lamp at P1-2 illuminates.

4.8.4.6 Resistors. To determine conformance to 3.5.5, the power supply of 4.8.4.1 shall be used to apply 24 ± 1.0 V dc across P1-19 and each of the pins specified in tables III or IV. Connect a multimeter of 4.8.4.1 in series with the individual resistor pins under test and the return lead of the power supply. verify the current flowing through each resistor is 37 ± 7.0 mA.

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4.8.4.7 Relay operations. Not applicable.

4.8.4.8 Transient voltage characteristics. To determine conformance to 3.5.7, the power supply shall be used to apply 1.0 ± 0.5 mA through a 2.4 K ± 10% ohm resistor to the positive and ground pins specified in 3.5.7. With a multimeter verify that the voltage from the positive to the ground pin is 39 ± 3 V dc.

4.8.4.9 Time delay and latching circuit (Class 2). To determine conformance to 3.5.8, utilizing power supply, apply 24 ± 6 V dc through an MS25237-387 test lamp to P1-28, and to a second MS25237-387 test lamp to P1-12. Connect the power source to P1-19 and the power source return to P1-15 for a 10 ± 2 second period. Verify that the two lamps do not illuminate. Connect the power source to P1-27 (for a 5 ± 1 second periods and the power source return to P1-15. Verify that the two test lamps illuminate and remain illuminated. Interrupt the power source and verify that the two test lamps extinguish.

4.8.4.10 Fire warning circuit. Not applicable.

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